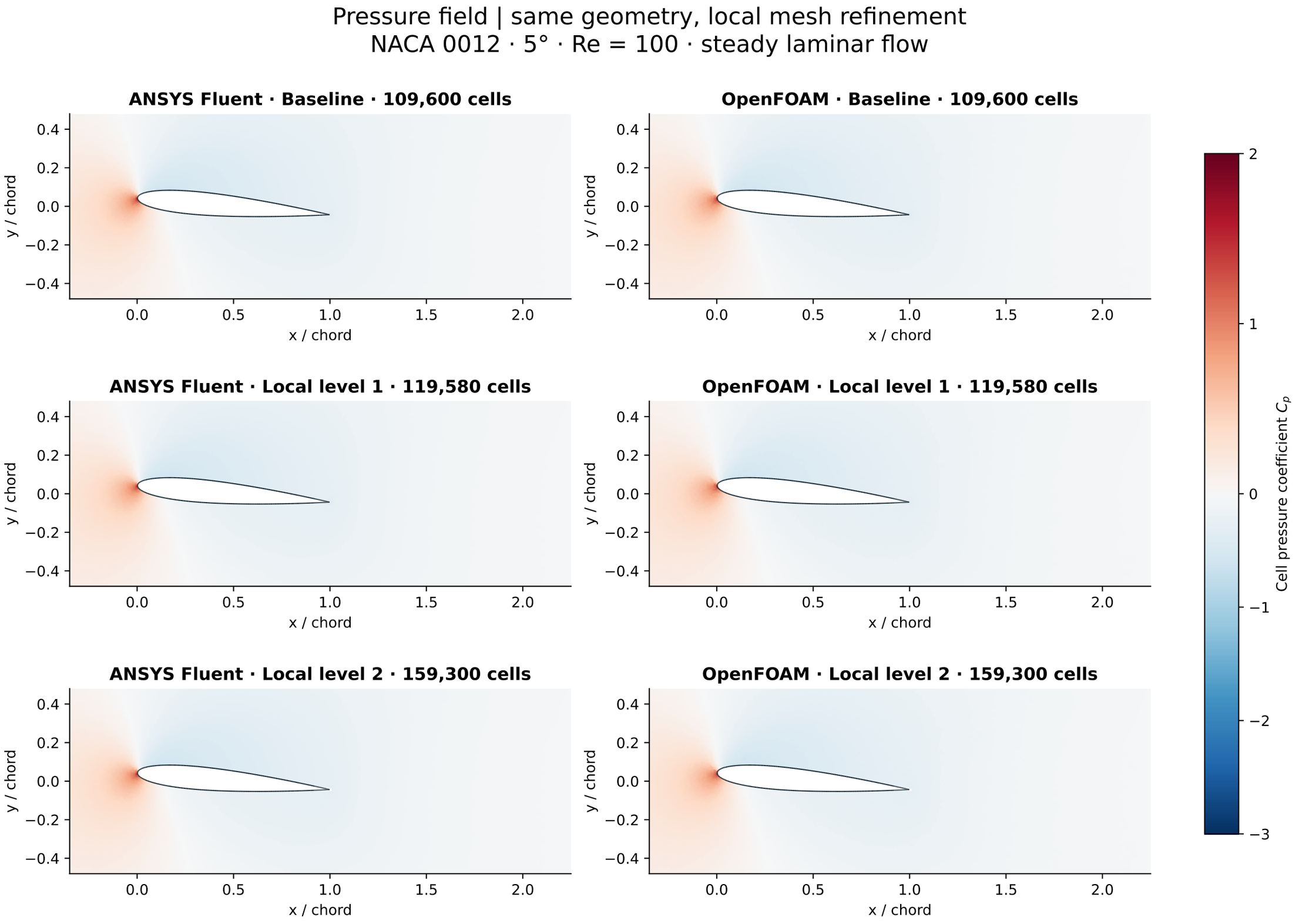

Pressure field | same geometry, local mesh refinement
NACA 0012 · 5° · Re = 100 · steady laminar flow
ANSYS Fluent · Baseline · 109,600 cells
OpenFOAM · Baseline · 109,600 cells
ANSYS Fluent · Local level 1 · 119,580 cells
OpenFOAM · Local level 1 · 119,580 cells
ANSYS Fluent · Local level 2 · 159,300 cells
OpenFOAM · Local level 2 · 159,300 cells
y / chord
x / chord
0.4
0.2
0.0
−0.2
−0.4
0.0
0.5
1.0
1.5
2.0
Cell pressure coefficient C_p
2
1
0
−1
−2
−3
Actual native cell values. Identical view and colour scale across all six cases; no temporal interpolation.
Numerical agreement is not physical validation or proof of mesh independence.